Roxana Petcu

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Scholar	scholar.google.com/citations?user=uVfPg-cAAAAJ

Bio

I am a PhD student in Information Retrieval at the University of Amsterdam. My research lies at the intersection of search and LLMs, with focus on training deep search agents with reinforcement learning (RL), LLM post-training, and RAG systems, often using reasoning-guided RL(VF), process reward modeling (PRMs), distillation, and self-supervision. More than that, I have experience in mechanistic interpretability for model alignment, agentic tool and skills execution, code generation, and representation learning.

Work Experience

Oct 2025 - Present — **Creator Fund**, London, UK
Lead Student Investment Partner
Guiding and evaluating technical feasibility of early-stage AI & Deep Tech startups in Europe

Nov 2022 - June 2023 — **Amazon Alexa**, Aachen Germany
Applied Science Intern
Project: Out of domain speech classifier for Alexa pre-training; 7% WER drop. amazon science
Supervisors: Volker Leutnant and Subhadeep Maji

Feb 2021 - July 2021 — **Amazon ATS**, Barcelona Spain
Applied Science Intern
Project: Forecasting Amazon shipments in south-west Europe for supply chain.
Supervisors: Thibault Gisselbrecht and Tim Forbes
Presented at *Amazon Machine Learning Conference* (AMLC)

Education

Sep 2023 - Present — **PhD University of Amsterdam, Information Retrieval Lab**
Complex information needs in retrieval systems
Supervisors: Evangelos Kanoulas, Maarten de Rijke

June 2025 - Aug 2025 — **Research visit, Johns Hopkins University**
RAG for request-guided Summarization of multilingual sources at the Human Language Technology Center of Excellence (HLTCOE).
Supervisors: Kevin Duh, Kenton Murray

Sep 2020 - Jul 2023 — **MSc Artificial Intelligence, University of Amsterdam**
Thesis: Efficient data selection using semantic graph structures for pre-training
Supervisor: Maarten de Rijke
Collaboration: Amazon Alexa

Sep 2017 - Jul 2020 — **BSc Computer Science, Vrije Universiteit**
Thesis: Approximate bayesian computation with differential evolution
Supervisor: Jakub Tomczak

Publications

2026 Local Proxies, Global Failures: Does Better Local Reasoning Improve Generation, or Do Models Strategize Differently?
Roxana Petcu, Evangelos Kanoulas, Maarten de Rijke
Under review

2026 SubSearch: Intermediate Rewards for Unsupervised Guided Reasoning in Complex Retrieval
Roxana Petcu, Evangelos Kanoulas, Maarten de Rijke
Under review
arxiv

2026 Balancing Exploration-Exploitation in Query Decomposition for RAG
Roxana Petcu, Kenton Murray, Daniel Khashabi, Evangelos Kanoulas, Maarten de Rijke, Dawn Lawrie and Kevin Duh
EACL
paper

2025 A Comprehensive Taxonomy of Negation for NLP and Neural Retrievers
Roxana Petcu, Samarth Bhargav, Maarten de Rijke and Evangelos Kanoulas
EMNLP
paper

2025 Reproducing and Extending Causal Insights Into Term Frequency Computation in Neural Rankers
Cile van Marken and **Roxana Petcu**
SIGIR-AP
paper

2025 Interpreting Multilingual and Document-Length Sensitive Relevance Computations in Neural Retrieval Models through Axiomatic Causal Interventions
Oliver Savolainen, Dur E Najaf Amjad and **Roxana Petcu**
SIGIR
paper

2025 Beyond Reproducibility: Advancing Zero-shot LLM Reranking Efficiency with Setwise Insertion
Jakub Podolak, Leon Peric, Mina Janicijevic and **Roxana Petcu**
SIGIR
paper

2024 Self-seeding and Multi-intent Self-instructing LLMs for Generating Intent-aware Information-Seeking dialogs
Roxana Petcu*, Arian Askari*, Chuan Meng, Mohammad Aliannejadi, Amin Abolghasemi, Evangelos Kanoulas, Suzan Verberne
NAACL
paper

2024 A Survey on Recent Advances in Conversational Data Generation
Roxana Petcu*, Heydar Soudani*, Evangelos Kanoulas, Faegheh Hasibi
CIKM, WWW
paper

2024 Leveraging Graph Structures to Detect Hallucinations in Large Language Models
Noa Nonkes*, Sergei Agaronian*, Evangelos Kanoulas, **Roxana Petcu**
ACL Workshop on Graph-based Methods for Natural Language Processing (TextGraphs-17)
paper

2023 Efficient data selection employing Semantic Similarity-based Graph Structures for model training
Roxana Petcu, Subhadeep Maji
ICML Workshop on Sampling and Optimization in Discrete Space (SODS)
paper, amazon science

2022 [Re] Replication Study of "Fairness and Bias in Online Selection"
Roxana Petcu, Pim Praat, Manolis Rerres, Jeroen Gerardus Samuel Wijnen
NeurIPS
paper

Talks

2026 Deep Research on Complex Queries
Roxana Petcu
Search Engines Amsterdam Meetup

2025 Causal Mediation in Mechanistic Interpretability
Roxana Petcu
UvA MSc advanced course on NLP

2025 Interpretability and Explainability in IR
Roxana Petcu, Maria Heuss
UvA MSc course on Information Retrieval

2024 Introduction to Transformers and the Attention Mechanism
Roxana Petcu, David Vos
UvA BSc AI, Information Studies and Computer Science
Location: Koninklijk Theater Tuschinski

2023, 2024 Tutorial on Recent Advances in Conversational Data Generation
Roxana Petcu*, Heydar Soudani*, Evangelos Kanoulas, Faegheh Hasibi
International Conference on Information and Knowledge Management (CIKM'23) and *International World Wide Web Conference (WWW'24)*

2021 Interpretability methods for Multivariate Time Series Forecasting
Roxana Petcu
Gheorghe Asachi Technical University, Faculty of Automation and Computers

Teaching

2023 Present Teaching Assistant, IR courses (MSc AI, UvA)
Supervised MSc student projects and theses in retrieval and RAG systems
Co-supervised industry-collaborative theses with partners including Elsevier and Vormedia

Academic Training and Schools

2025 NLDL 2025 (Northern Lights Deep Learning Winter School)

2024 ESSIR 2024 (European Summer School on Information Retrieval)

2024 GeMMS 2024 (Generative Modeling Summer School)

Organizational Roles

2026 SCAI 2026 (Workshop on Search-Oriented Conversational AI) at SIGIR 2026

2025 Workshop on Conversational Search for Complex Information Needs at ECIR 2026

2024 ESSIR 2024 (European Summer School on Information Retrieval)

2022 Mentor at NeurIPS 2022 Education Outreach Day

Sept 2024 - Now IRLab, University of Amsterdam
Organized cross-institute Mechanistic Interpretability reading group, UvA